NBA Shot Analysis

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2026-04-12

Column Types

The data set of nbaShots05_06.csv is actively tracking each shot attempt throughout the regular season by players in an instance form. There are several things captured in the dataset, notated by the column names.

```
library(readxl)
library(kableExtra)
```

```
## Warning: package 'kableExtra' was built under R version 4.5.3
```

```
options(max.print = 500)

df <- data.frame(read_excel("C:/Users/Rober/Desktop/Python Projects/NBA Shot Analysis QMB 3311/analysis/


kable(df)
```

Column.Name	Column.Description
GRID_TYPE	Database/query the data was extracted from.
GAME_ID	ID that the NBA assigns to games, unique per game.
GAME_EVENT_ID	ID for event in the gameID: fouls, shots, blocks, etc.
PLAYER_ID	The unique player ID of the shot
PLAYER_NAME	the corresponding player name to the ID for the record
TEAM_ID	TeamID that the player is on (CAN be different with trades)
TEAM_NAME	The corresponding team name to the Team ID
PERIOD	the quarter of the game that the record is in
MINUTES_REMAINING	the time left in the quarter when the record is made
SECONDS_REMAINING	the seconds left on the clock when the record is made
EVENT_TYPE	What event was specifically tracked, Miss or Make
ACTION_TYPE	The type of event, in this case, specific type of shot.
SHOT_TYPE	The label given to the scoring type of shot, 2pt or 3pt.
SHOT_ZONE_BASIC	the basic area where the shot was taken from
SHOT_ZONE_AREA	More granular descripter of the location within the basic zone
SHOT_ZONE_RANGE	Range to the rim of the shot in 8ft increments
SHOT_DISTANCE	Specific distance of the shot
LOC_X	X coordinate of the shooter's position on the court
LOC_Y	Y coordinate of the shooter's position on the court
SHOT_ATTEMPTED_FLAG	flag variable indicating that the event is a shot attempt
SHOT_MADE_FLAG	flag variable indicating the shot was Made(1) or Missed(0)

Column.Name	Column.Description
GAME_DATE	the date of the game in Y-M-D format
HTM	The home team abbreviation
VTM	The away team abbreviation

Summary Statistics

Using the summary statistics, I decided to view the death of the mid-range shot from when it still had value in 2005-2006. To process this in summary statistics, I specifically used the mean. I ran this chunk in python to extract an expected points value from each 'SHOT_ZONE_BASIC'.

```
# ---- Summary Statistics, Ex Point p shot ----

zone_eps = (
    shots_data.groupby("SHOT_ZONE_BASIC")
    .agg(
        attempts=("SHOT_ATTEMPTED_FLAG", "sum"),
        makes=("SHOT_MADE_FLAG", "sum"),
        avg_point_value=("POINT_VALUE", "mean"),
    )
    .reset_index()
)

zone_eps["fg_pct"] = zone_eps["makes"] / zone_eps["attempts"]
zone_eps["eps"] = zone_eps["fg_pct"] * zone_eps["avg_point_value"]
zone_eps = zone_eps.sort_values("eps", ascending=False)

print(zone_eps.to_string(index=False, float_format="%.3f"),'\n') # summary of the EPS calculated
```

```
##       SHOT_ZONE_BASIC  attempts  makes  avg_point_value  fg_pct   eps
##       Restricted Area     63017  37304            2.000   0.592 1.184
##        Right Corner 3      5409   2123            2.996   0.392 1.176
##         Left Corner 3      5668   2184            3.000   0.385 1.156
##     Above the Break 3     27816   9752            3.000   0.351 1.052
## In The Paint (Non-RA)     25107  10167            2.000   0.405 0.810
##             Mid-Range     66926  26627            2.001   0.398 0.796
##             Backcourt       371      9            2.997   0.024 0.073
```

This is visualized here:

The mid range 2 point shot is classified as the Mid Range and the In The Paint (Non-RA). Evidently, they provide much less expected return, likely due to closer defenders, bunched formations, and more complex shot angles. In contrast, the higher likllihood of the near the rim dunks and layups significantly increased its value in comparison. While still only a 60% clip, the near rim shot is the vastly superior 2 pointer.

Even in 2005, the lower rate at which 3 pointers were made did not devalue the 3 point range below a 2 point shot. The data clearly shows the earned value and benefit of taking the longer range shots less of the time. However, there is a discrepancy to this hypothesis that is explored in the Regression Statistics.

Regression Analysis

In the regression analysis, I attempted to identify a link between a team's aggregate expected points (from their shot types and shooting percentages) with their season ending win totals. This was done via the

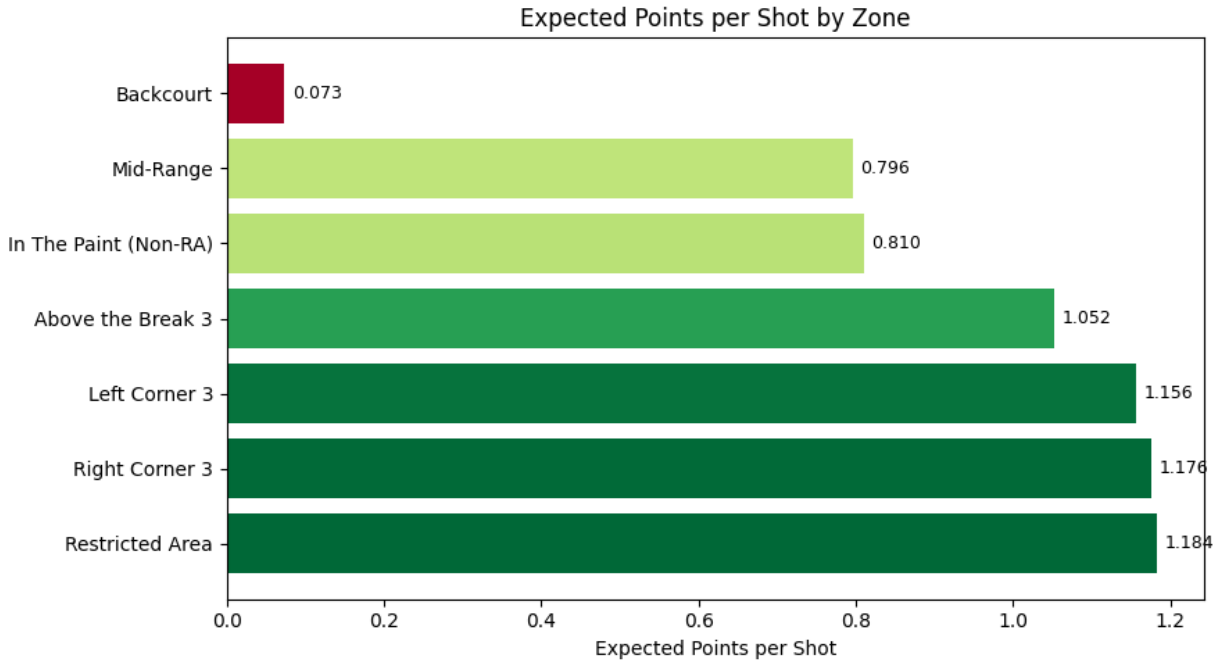

Figure 1: *Expected Points per Shot in Zones*

following code chunk:

```
# ---- Regressing, predicted wins from aggregated team eps ----

team_season_eps = (
    shots_data.groupby("TEAM_NAME")
    .agg(
        attempts=("SHOT_ATTEMPTED_FLAG", "sum"),
        makes=("SHOT_MADE_FLAG", "sum"),
        avg_point_value=("POINT_VALUE", "mean"),
        avg_distance=("SHOT_DISTANCE", "mean"),
    )
    .reset_index()
)
# team specific aggregated eps via fg%*avg point value of shots
team_season_eps["fg_pct"] = team_season_eps["makes"] / team_season_eps["attempts"]
team_season_eps["eps"] = team_season_eps["fg_pct"] * team_season_eps["avg_point_value"]
```

This was then linked to their season record:

```
regression_df = team_season_eps.merge(records, on="TEAM_NAME", how="inner")

print(
    regression_df[["TEAM_NAME", "Wins", "Losses", "fg_pct", "eps", "attempts", "avg_distance"]]
    .sort_values("Wins", ascending=False)
    .to_string(index=False, float_format="%.3f")
)
```

```
##                          TEAM_NAME  Wins  Losses  fg_pct   eps  attempts  avg_distance
##                     Detroit Pistons    64      18   0.455 1.010      6558        12.818
##                   San Antonio Spurs    63      19   0.472 1.045      6342        11.422
##                    Dallas Mavericks    60      22   0.462 1.006      6375        11.485
##                        Phoenix Suns    54      28   0.479 1.097      7167        13.985
##                          Miami Heat    52      30   0.478 1.065      6355        11.041
##                  Cleveland Cavaliers    50      32   0.454 1.011      6412        11.495
##                      New Jersey Nets    49      33   0.440 0.980      6317        12.531
##                    Memphis Grizzlies    49      33   0.448 1.012      6125        11.663
##                 Los Angeles Clippers    47      35   0.465 0.990      6443        11.509
##                   Los Angeles Lakers    45      37   0.453 1.014      6607        12.450
##                       Denver Nuggets    44      38   0.461 0.997      6672        11.061
##                     Sacramento Kings    44      38   0.454 1.007      6500        12.897
##                   Washington Wizards    42      40   0.447 0.988      6656        12.154
##                            Utah Jazz    41      41   0.442 0.950      6207        11.341
##                        Chicago Bulls    41      41   0.446 0.990      6737        12.626
##                       Indiana Pacers    41      41   0.444 0.997      6274        12.430
##                      Milwaukee Bucks    40      42   0.453 0.999      6566        11.066
## New Orleans/Oklahoma City Hornets    38      44   0.440 0.941      6428        10.853
##                   Philadelphia 76ers    38      44   0.458 0.989      6546        10.709
##                         Orlando Magic    36      46   0.472 1.005      6167        10.545
##                  Seattle SuperSonics    35      47   0.459 1.028      6711        12.146
##                      Houston Rockets    34      48   0.433 0.964      6252        12.230
##                Golden State Warriors    34      48   0.433 0.983      6785        12.671
##               Minnesota Timberwolves    33      49   0.456 0.979      6340        12.084
##                       Boston Celtics    33      49   0.467 1.029      6322        11.564
##                      Toronto Raptors    27      55   0.454 1.018      6639        12.488
##                    Charlotte Bobcats    26      56   0.433 0.945      6843        11.576
##                        Atlanta Hawks    26      56   0.454 0.988      6496        10.996
##                      New York Knicks    23      59   0.455 0.974      6205         9.724
##               Portland Trail Blazers    21      61   0.445 0.965      6267        11.579
```

The regression model yielded some interesting results as it did not directly match the hypothesis. Instead it opened further questions. There was a correlation between the wins and shot efficiency; however some teams that used a mid of the table shot efficiency had the most regular season wins: The Pistons, Mavericks, and the Spurs. The Phoenix Suns used a high rate of three point shots to generate their EPS and the Miami Heat used a mix of play around the hoop and threes to keep the second highest value. As shown in Figure 2.

External Factors

There are several external factors not considered in this single data file. These include statistics such as rebounds, free throw percentages, or even the number of free throws per team. The last one is significant because two point attempts are much more likely to result in a free throw than a three pointer. The increase of the expected points value of the 2 point shots are not included here. I would need the foul event data and be able to connect between the shot event and the foul event to make the connection. I did make a visual showing each team's shot type rates, and it showed higher 2 point attempt rates for some of the outperforming teams, lending some credence to the free throw hypothesis. As shown in Figure 3.

Other Notes

While there is not a significant correlation between the EPS taken by each team and the regular season wins, the methodology is somewhat comforted by the Miami Heat winning the NBA finals this season with the second highest EPS value.

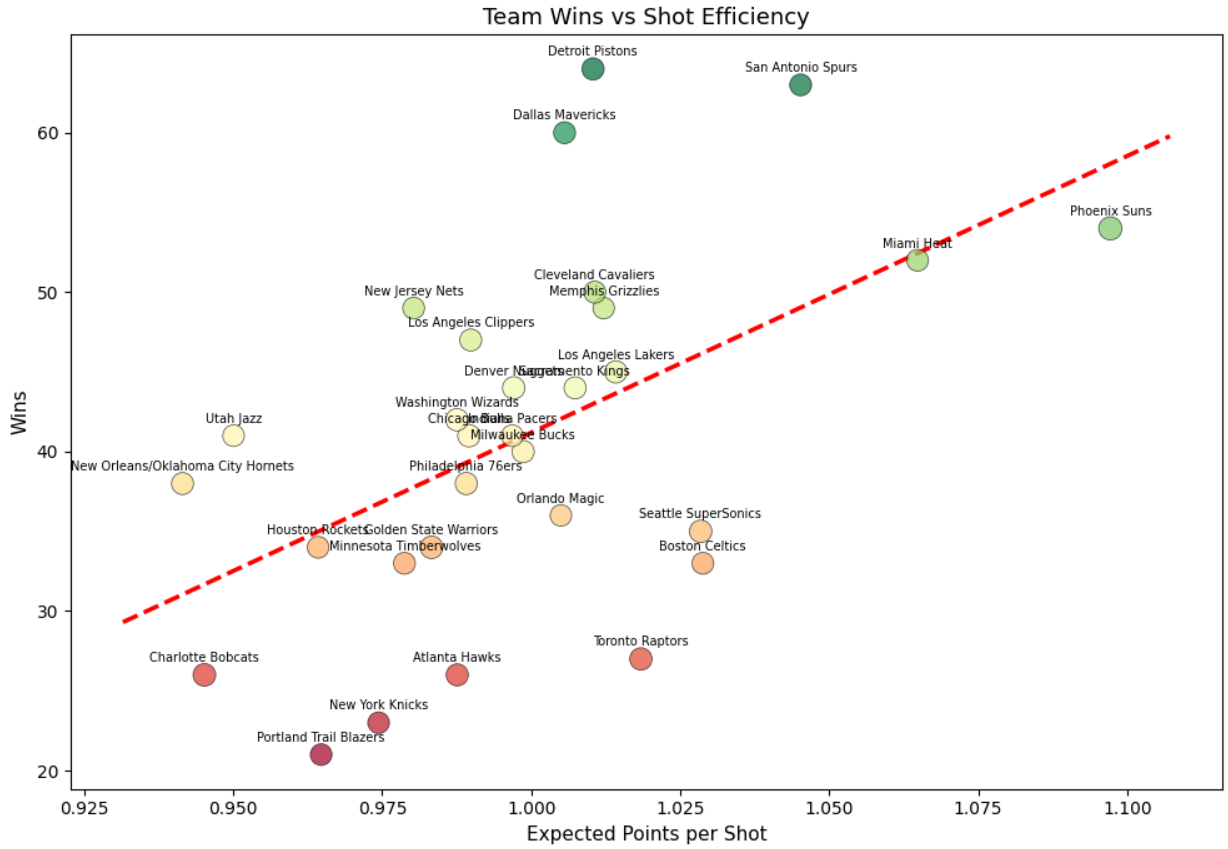

Figure 2: *Regression Model*

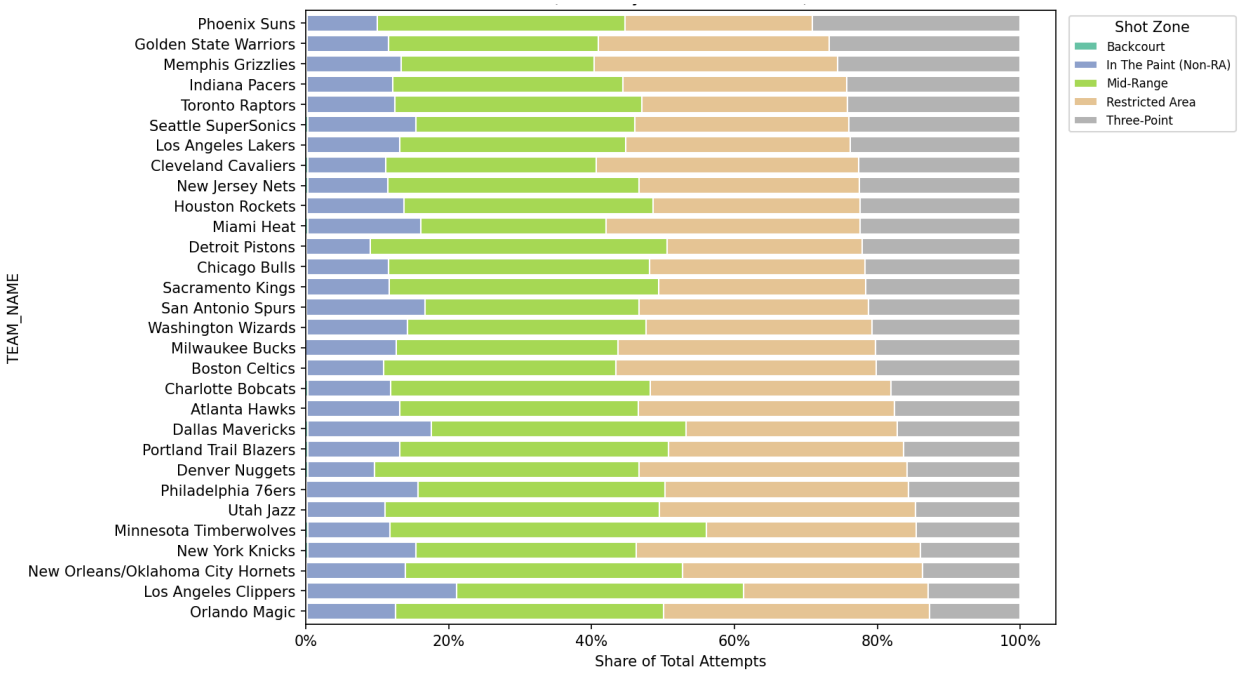

Figure 3: *Shot Type by Team*